

⚡ WD002 Week 3 — English Worksheet

Topic: Events — Reacting to the User · **Course:** Web Dev Basics: JavaScript · **Time:** about 45 minutes

This week your page **listens**. An *event* is something the user does — a click, a hover, a key press.

`addEventListener("click", ...)` runs a function when that event happens. You will build a score counter that goes up, down, and resets.

1 · Predict 🧠

Read each piece of code. Before you run it, write what you think will happen.

```
<button id="btn">Tap</button>

<script>
  document.querySelector("#btn").addEventListener("click", function() {
    console.log("tapped");
  });
</script>
```

What prints when you click the button three times?


```
<span id="score">0</span>
<button id="up">+1</button>

<script>
  let score = 0;
  document.querySelector("#up").addEventListener("click", function() {
    score = score + 1;
    document.querySelector("#score").textContent = score;
  });
</script>
```

You click **+1** four times. What number is on the screen?


```
<button id="reset">Reset</button>

<script>
  let score = 10;
  document.querySelector("#reset").addEventListener("click", function() {
    score = 0;
    document.querySelector("#score").textContent = score;
  });
</script>
```

The score starts at 10. After clicking reset, what is it?


```
<button id="down">-1</button>

<script>
  let score = 2;
  document.querySelector("#down").addEventListener("click", function() {
    score = score - 1;
    document.querySelector("#score").textContent = score;
  });
</script>
```

You click **-1** five times starting from 2. What number shows? Is anything strange about it?

2 · Spot the Bug 🐛

Each block was meant to do something, but it is broken. Read what it is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — Clicking the button should add 1 to the score on screen. Right now the number on screen never changes, even though the click happens.

```
<span id="score">0</span>
<button id="up">+1</button>

<script>
  let score = 0;
  document.querySelector("#up").addEventListener("click", function() {
    score = score + 1;
  });
</script>
```

Hint: the variable goes up, but nobody updates the screen. One line is missing.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This should listen for a **click**. Right now clicking the button does nothing.

```
<button id="go">Go</button>

<script>
  document.querySelector("#go").addEventListener("clik", function() {
    console.log("go!");
  });
</script>
```

Hint: the event name is misspelled, so the browser is listening for an event that never happens.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The score should never drop below 0. Right now clicking **-1** past zero gives negative numbers.

```
<button id="down">-1</button>

<script>
  let score = 0;
  document.querySelector("#down").addEventListener("click", function() {
    score = score - 1;
    document.querySelector("#score").textContent = score;
  });
</script>
```

Hint: before subtracting, check whether the score is already 0. Use an **if**.

Write the fixed code:

Why was it wrong? Why does your fix work?

+1

-1,

reset

- +1

-1

reset

```
if )
```

+5

photo or video

I used `addEventListener` to ...

The reset button works by ...

If I had more time, I would ...

4 · Record Your Walkthrough

Take a video on your phone (or a parent's phone) while your counter runs. Talk through it like you are teaching someone who has never seen it. Try to use these words: **event**, **listener**, **click**, **score**, **reset**.

1. Show the counter at 0.
2. Click **+1** a few times, then **-1** , then **reset** .
3. Read one listener out loud and say which button it belongs to.
4. Show what happens when you try to go below 0.

Write what you will say in your video. Plan it here before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.